Arun Jayaraman

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Summary

Senior scientist with 4 years of experience in translational bioinformatics, leveraging multi-omics and deep profiling to guide clinical trial design and de-risk drug development pipelines at AstraZeneca and AbbVie. Proven expertise in identifying and assessing novel predictive biomarkers in immunology and respiratory disease, developing automated analysis tools, and translating complex omics insights into actionable clinical and business strategies.

Work Experience

Senior Scientist — *Jan 2023 – Present*
AstraZeneca — *Gaithersburg, MD*

- De-risked a multi-million dollar Phase II trial by leading multi-omic analysis of rheumatoid arthritis (RA) cohorts, delivering a key biomarker strategy that informed patient selection and investment decisions.
- Demonstrated concordance between type 2 inflammation markers and nasal expression using RNA-seq and Olink data from 245 patients, validating non-invasive nasal sampling as a proxy for lower airway biology.
- Identified new biomarker opportunities by assessing multiple autoantibody profiling platforms in an autoimmune cross-disease cohort.
- Informed business development strategy by providing computational analyses across atopic dermatitis, inflammatory bowel disease, and Sjogren's disease.
- Accelerated translational research by engineering a centralized repository of pre-computed gene, protein, and signature statistics with an integrated R Shiny interface, automating figure generation and improving internal data access.

Associate Scientist II — *Jan 2022 – Jan 2023*
AbbVie Genomics Research Center — *North Chicago, IL*

- Identified novel disease drivers in therapeutic non-responders using integrated single-cell and bulk transcriptomic analyses of rheumatologic cohorts.
- Discovered a pathogenic synovial fibroblast population enriched in RA therapeutic non-responders.
- Leveraged linear mixed-effects models for longitudinal analysis of upadacitinib-treated arthritis patients, validating drug-specific signatures and informing drug repurposing strategy.

Graduate Researcher — *Sept 2020 – Dec 2021*
Emory University, The Davis Lab — *Atlanta, GA*

- Pioneered scRNA-seq analysis of Cardiac Progenitor Cells (CPCs) to identify populations driving cardiac repair (Published in *J. Cardiovasc. Dev. Dis.*).
- Defined age-related shifts in CPC transcriptional profiles by analyzing competing endogenous RNA networks (Published in *Genomics*).
- Collaborated with Yale School of Medicine to detect transplant rejection via T cell response in mice (Published in *Am. J. Transplant.*).

AI/ML-Driven Drug Discovery and Cheminformatics — *Summer 2020*
AbbVie Innovation Center — *Urbana, IL*

- Leveraged conditional generative models (VAE, GAN, RL) for the *de novo* design and generation of novel molecules.
- Developed a hierarchical multi-output neural network to predict optimal chemical reaction conditions, improving efficiency in high-throughput screening.

Education

Georgia Institute of Technology
M.S. Biomedical Engineering, Dec 2021

University of Illinois at Urbana-Champaign
B.S. Chemical Engineering, May 2020
With Distinction, Dean's List

Technical Skills

Programming
R, Python, JavaScript, Bash, SQL

Computational Pipelines
Nextflow, Snakemake

Machine Learning
Scikit-learn, TensorFlow, PyTorch

Data Visualization
Tidyverse (ggplot2), R Shiny, Matplotlib

Bioinformatics

Alignment:	STAR, Salmon, Cell Ranger, Space Ranger
Differential Exp.:	DESeq2, edgeR, limma, mixed-effects models (lme4, dream)
Signature Analysis:	GSVA, singscore, WGCNA, AUCell
Single Cell:	Seurat, Scanpy, Harmony, Monocle 3

Publications & Presentations

Peer-Reviewed Publications

- Korutla L, Hoffman J.R., ... **Jayaraman, A.**, et al. (2024). Circulating T cell specific extracellular vesicle profiles in cardiac allograft acute cellular rejection. *Am. J. Transplant*.
- Hoffman, J., **Jayaraman, A.**, Bheri, S., & Davis, M. (2022). Single cell RNA-sequencing reveals distinct cardiac-derived stromal cell subpopulations. *J. Cardiovasc. Dev. Dis.*
- Gunasekaran, M., Mishra, R., ... **Jayaraman, A.**, et al. (2022). Comparative efficacy and mechanism of action of cardiac progenitor cells after cardiac injury. *iScience*.
- Hoffman, J., Park, H., ... **Jayaraman, A.**, & Davis, M. (2022). Comparative computational RNA analysis of cardiac-derived progenitor cells and their extracellular vesicles. *Genomics*.
- Park, J., **Jayaraman, A.**, Schrader, A. W., Hwang, G. W., & Han, H. (2020). Controllable modulation of precursor reactivity for synthesis of high-quality quantum dots. *Nature Communications*.
- Park, J., **Jayaraman, A.**, Wang, X., Zhao, J., & Han, H. (2020). Nanocrystal precursor incorporating separated mechanisms for nucleation and growth. *ACS Nano*.
- Capece, M., Borchardt, C., & **Jayaraman, A.** (2020). Improving the effectiveness of the Comil as a dry-coating process: enabling direct compaction for high drug loading formulations. *Powder Technology*.

Conference Abstracts and Presentations

- "Prevalence of IFN high rheumatoid arthritis in early and established RA patients" (2025). Abstract in *Ann Rheum Dis*.
- "Elevated Serum Peptidylarginine Deiminase 4 (PAD4) Levels in Anti-CCP Antibody Positive At-Risk Individuals with Arthralgia Who Progress to Rheumatoid Arthritis" (2025). Poster presentation, ACR Convergence. Abstract in *Ann Rheum Dis*.
- "Nasal sampling identifies disease relevant signals in real-world airways disease cohort" (2024). Oral presentation, European Respiratory Society Congress. Abstract in *European Respiratory Journal*.
- "The NOVELTY nasal transcriptome identifies mechanisms of asthma and COPD" (2024). Poster presentation, ERS Congress. Abstract in *European Respiratory Journal*.
- "The nasal transcriptome reflects clinically relevant disease activity in NOVELTY, with a different signal in asthma and COPD" (2024). Poster presentation, ERS Congress. Abstract in *European Respiratory Journal*.